Katelyn Gan

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EDUCATION

Massachusetts Institute of Technology B.S. Mathematics and Artificial Intelligence, May 2028 (Expected)

Relevant Coursework: Probability and Random Variables, Fundamentals of Statistics, Design and Analysis of Algorithms, Introduction to Machine Learning, Algebra I

WORK & RESEARCH EXPERIENCE

Backend Software Engineering Intern | *Thinkstruct, Boston, MA* June 2026 – August 2026

Built distributed AI scraping and ingestion pipelines for 5M+ patent documents and scaled backend concurrency via Postgres migration and an async LLM generation system (S3), successfully eliminating request timeouts.

MIT Undergraduate Researcher | *PI: Kaiming He, CS and Artificial Intelligence Lab (CSAIL)* Dec. 2025 – Present

- Developed 10+ multimodal benchmarks (30K+ samples/task) and a resumable, diffusion data pipeline on SLURM GPU clusters, fine-tuning models via LoRA and deploying LLM-as-judge verifiers to filter data at scale.
- Optimized Vision Transformers on the Abstraction and Reasoning Corpus using self-supervised learning, rotary positional embeddings, and test-time training on Cloud TPU clusters.

Director of Recruitment | *MIT Pi Beta Phi, Cambridge, MA* May 2026 – Present

Directed end-to-end recruitment logistics, budgets, candidate pipelines, and committee teams for a 100+ member chapter.

MIT PRIMES-USA Math Researcher | *Mentor: Yunseo Choi, Harvard University* Jan. 2024 – Feb. 2025

Simulated lattice-based combinatorial games in Python and proved open conjectures related to game-theoretic behavior.

SELECTED PUBLICATIONS

Xiaoman Ding, Keya Hu, Katelyn Gan, Victor Yin, Kaiming He, "Natural Image Pretraining Improves Abstract Reasoning", Accepted to the European Conference on Computer Vision (ECCV), 2026

Yunseo Choi, Katelyn Gan, Andrew Li, Tiffany Zhu, "*Set Partitions that Require a Maximum Number of Sorts Through the aba-avoiding Stack,*" Enumerative Combinatorics and Applications, Volume 5:1, Article S2R1, 2025.

Yunseo Choi and Katelyn Gan, "*Ungar Games on the Young-Fibonacci and the Order Ideals of Shifted Staircase Lattices,*" submitted for publication, oral presentation at Joint Mathematics Meetings, Seattle, WA, 2025.

Christopher Bao, Giulio Cerbai, Yunseo Choi, Katelyn Gan, Owen Zhang, "*On a Conjecture on Pattern-avoiding Machines,*" Annals of Combinatorics, 2024. Also presented at Joint Mathematics Meetings, San Francisco, CA, 2024.

Katelyn Gan and Vinesh Maguire Rajpaul, "*A Computer Vision Approach to Radial Velocity Extraction for Exoplanet Detection,*" 2023 IEEE MIT URTC, Cambridge, MA, USA, 2023.

OTHER INDEPENDENT PROJECTS

arXiv AI Daily: Automated Research Intelligence System (GitHub)

Engineered a scalable pipeline to scrape, embed, and rank arXiv cs.AI papers using FAISS and OpenAI embeddings.

A Toy Language Model: GPT-Style Transformer

Built a PyTorch GPT-style model featuring multi-head self-attention and retrieval-augmented generation via FAISS.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Java, MATLAB, JavaScript

Frameworks: PyTorch, JAX, OpenCV, FAISS, React, Node.js, Firebase

Infra & Tools: GCP, AWS, Docker, SLURM, TPUs (v5/v6), NVIDIA GPUs, Linux, Git, LaTeX

SELECTED AWARDS & HONORS

MIT Pozen Fellowship | Jane Street WiSE | Discover Citadel | D. E. Shaw Connect | Jump Trading Early Access | USAAIO Gold | USAMO & USAPhO Honorable Mention | ISEF 4th Place Grand Prize | OpenCV AI Competition, 2nd Place